

Wearable Technologies and Applications (Wearable Informatics)

Winfree

Lecture 3

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

What's Inside?



INF632
(EE499/EE599)

Winfree

Senses

What's Inside?

Define

Defined: Haptic
Defined: (Opt)ics
Defined: Haptics

Haptic Exploration

What's Inside?
Exploratory Procedures
Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices
Tactile Device Challenges
Kinesthetic Device
Challenges
Workspace vs Force
Haptic Wearables?

Readings

Thursday

Winfree

-
- A simple stick figure sketch of a person in a dynamic, walking or running pose. The figure is composed of straight lines for the limbs and torso, with a large, rounded head. The right leg is extended forward, and the left leg is bent, suggesting movement. The arms are also in motion, with the right arm bent and the left arm extended back. The overall impression is one of active, purposeful movement.

Five Senses, Which is Most Important to You?

INF632
(EE499/EE599)

Winfree

Sight



- ▶ Centralized
- ▶ Broad
- ▶ Passive
- ▶ Cognitive

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

Winfree

Senses

What's Inside?

Define

Haptic Exploration

Haptic Devices

Readings

Sight



- ▶ Centralized
- ▶ Broad
- ▶ Passive
- ▶ Cognitive



Touch



- ▶ Distributed
- ▶ Narrow
- ▶ Interactive
- ▶ Fundamental

What is Haptics?

Definition

haptic

adjective

1. of or relating to the sense of touch, in particular relating to the perception and manipulation of objects using the senses of touch and proprioception.

origin

1. late 19th century: from Greek *haptikos* 'able to touch or grasp,' from *haptēin* 'fasten.'

INF632
(EE499/EE599)

Winfree

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

Winfree

Senses

Defined: (Opt)ics

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ≡ ↺ 🔍 ↻

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

Definition

Haptics

- ▶ the science and technology of touch

What's Inside?



INF632
(EE499/EE599)

Winfree

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

What's Inside?



INF632
(EE499/EE599)

Winfree

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

Exploratory Procedures

INF632
(EE499/EE599)

Winfree

Lateral Motion
Texture



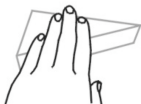
Pressure
Hardness



Enclosure
Global shape/Volume



Static Contact
Temperature



Unsupported Holding
Weight



Contour Following
Shape



Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

Types of Sensing

Cutaneous / Tactile
Stems from skin contact



Kinesthetic / Proprioceptive
Stems from body movement

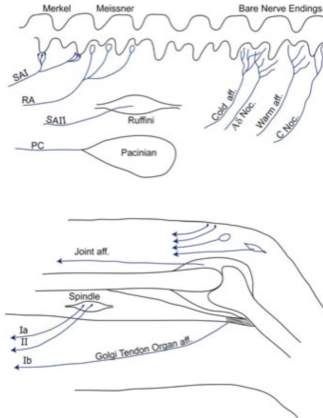


Illustration by A. M. Okamura.

INF632
(EE499/EE599)

Winfree

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

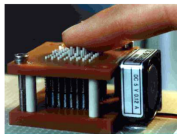
Readings

Thursday

Tactile and Proprioceptive Devices

INF632
(EE499/EE599)

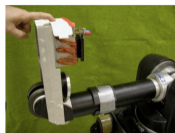
Winfree



Tactile Devices

Stimulate skin to create contact sensations

Haptic Interfaces



Hybrid Devices

Attempt to combine tactile and kinesthetic feedback



Kinesthetic Devices

Apply forces to guide or inhibit body movement

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

Tactile Device Challenges

INF632
(EE499/EE599)

Winfree

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

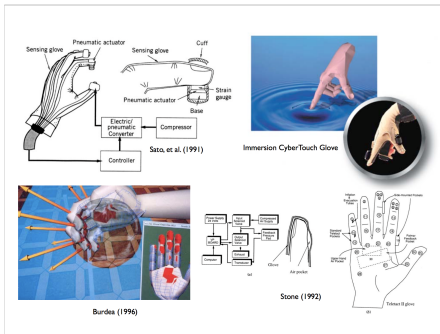
Kinesthetic Device
Challenges

Workspace vs Force

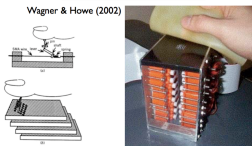
Haptic Wearables?

Readings

Thursday



Wagner & Howe (2002)



Kontarinis, et al. (1995)

- ▶ Stimulation density needs to be high.
- ▶ Multimodal sensations are very hard to overlay.
- ▶ High complexity and weight diminish portability.
- ▶ Passive touch does not feel real.
- ▶ Tactile device design is difficult!

Kinesthetic Device Challenges

INF632
(EE499/EE599)

Winfree



Immersion CyberGrasp Glove



- ▶ Competing goals of high stiffness and low mass.
- ▶ Force feedback feels soft - “Nerfworld”.
- ▶ Point based interactions are overly simple.
- ▶ High quality devices are expensive.
- ▶ Limited workspace.
- ▶ Constrained to a desk!

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive Devices

Tactile Device Challenges

Kinesthetic Device Challenges

Workspace vs Force

Haptic Wearables?

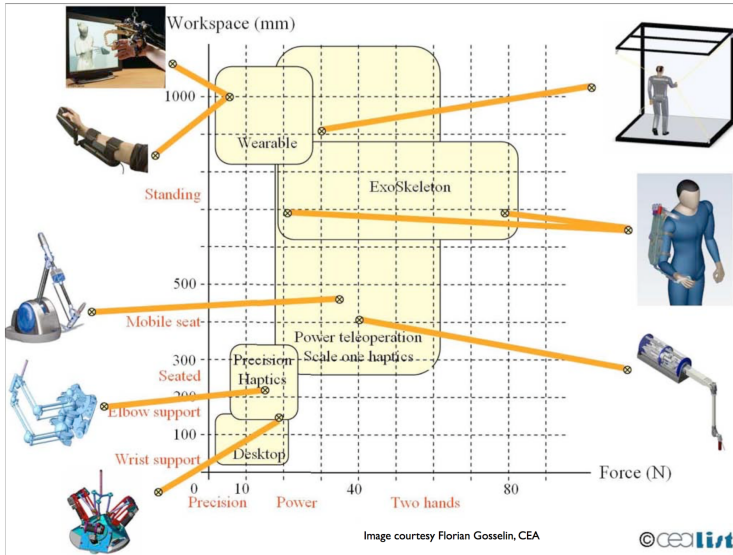
Readings

Thursday

Workspace vs Force

INF632
(EE499/EE599)

Winfree



Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive Devices

Tactile Device Challenges

Kinesthetic Device Challenges

Workspace vs Force

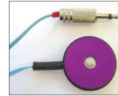
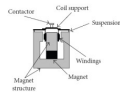
Haptic Wearables?

Readings

Thursday

But what can you do with haptic wearables?

- ▶ Vibration (eccentric mass motor or tactor)
- ▶ Skin stretch
- ▶ Kinesthetic force
- ▶ Inertial changes



INF632
(EE499/EE599)

Winfree

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive Devices

Tactile Device Challenges

Kinesthetic Device Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

- ▶ OPTIONAL - Embodied Auxetic Intelligence in a Glove-Type Wearable Haptic Interface Connecting Humans.pdf
- ▶ OPTIONAL - FiDTouch - A 3D Wearable Haptic Display for the Finger Pad.pdf
- ▶ OPTIONAL - Haptics in Teleoperated Medical Interventions - Force Measurement, Haptic Interfaces and Their Influence on Users Performance.pdf
- ▶ OPTIONAL - Music Tactalizer - A Wearable Haptic Music Player with Multi-Feature Audio-Tactile Rendering.pdf
- ▶ OPTIONAL - Predicting the Future with Wearable Technology.pdf
- ▶ OPTIONAL - The Future of Wearable Technologies and Remote Monitoring in Health Care.pdf
- ▶ OPTIONAL - Ultralight Soft Wearable Haptic Interface with Shear-Normal-Vibration Feedback.pdf

Senses

What's Inside?

Define

Defined: Haptic

Defined: (Opt)ics

Defined: Haptics

Haptic Exploration

What's Inside?

Exploratory Procedures

Types of Sensing

Haptic Devices

Tactile and Proprioceptive
Devices

Tactile Device Challenges

Kinesthetic Device
Challenges

Workspace vs Force

Haptic Wearables?

Readings

Thursday

- ▶ OPTIONAL - Embodied Auxetic Intelligence in a Glove-Type Wearable Haptic Interface Connecting Humans.pdf
- ▶ ...
- ▶ OPTIONAL - Ultralight Soft Wearable Haptic Interface with Shear-Normal-Vibration Feedback.pdf
- ▶ **TO DISCUSS - Haptic Perception and Its Relation to Action.pdf**
- ▶ **TO DISCUSS - Haptic Rendering Introductory Concepts.pdf**
- ▶ **TO DISCUSS - The future of wearable technologies.pdf**